THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SOCIAL SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 2: **Understanding the Weather** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A city shows a rise in mean night temperatures over decades. A likely cause is 1

(a) urban heat island and climate change **(b)** more lakes
(c) more forests **(d)** fewer buildings

2. Which of the following measure could reduce loss of life during cyclones? 1

(a) ignoring signs **(b)** no information
(c) early warnings and evacuation **(d)** late warnings

3. If humidity rises at constant temperature and the air nears 100 per cent, the likely result is 1

(a) no vapour **(b)** cloud or fog formation
(c) drier air **(d)** clear skies

4. Why is a rain gauge placed in an open area? 1

(a) to avoid trees and buildings blocking rain
(b) for wind
(c) for noise
(d) for shade

5. How does the Sun's heating of the Earth drive the water cycle? 1

(a) stops water **(b)** evaporates water and moves the air
(c) makes ice **(d)** makes wind stop

6. Weather data for a month shows 0 mm rain, high temperatures and low humidity. The likely season is 1

(a) monsoon **(b)** cyclone **(c)** dry summer **(d)** winter fog

7. Why is a thermometer placed in a shaded, ventilated box? 1

(a) to cool it
(b) to hide it
(c) to measure air temperature without direct sunlight
(d) to heat it

8. The relative humidity is 50 per cent. If the temperature falls, relative humidity 1

(a) stays the same **(b)** falls **(c)** becomes zero **(d)** rises

9. **Assertion (A):** Normal atmospheric pressure at sea level is about 1013 millibars. 1
Reason (R): A barometer is used to measure rainfall.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Wet clothes dry faster when humidity is high. 1
Reason (R): High humidity means the air already holds a large amount of water vapour.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is a barometer useful to sailors? 2

12. How can citizens use IMD alerts? 2

13. Why is relative humidity high at night? 2

14. Why does cloud cover reduce daily range? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Examine why the Himalayas have snow while the plains have rain. 3

16. Examine the effect of vegetation on local humidity and temperature. 3

17. Examine how clouds influence temperature in a day. 3

OR

Evaluate the accuracy of traditional weather signs. 3

18. Explain how urbanisation affects local weather. 3

OR

Evaluate the importance of weather instruments in schools. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** On a sunny afternoon, the air over land near the Gujarat coast becomes warmer, and its pressure falls. The air over the cooler sea has higher pressure. Winds start to blow from the sea towards the land. A wind sock at the nearby airstrip points inland. 5
(i) From which area does the wind blow here?
(ii) Where is the pressure lower?
(iii) Which instrument at the airstrip shows wind direction?
(iv) Explain why wind blows from the sea to the land.

20. Compute and interpret mean temperature and range for a week of given data and explain the weather pattern. 5

OR

Evaluate how weather knowledge can reduce risks for farmers. 5

END OF QUESTION PAPER — Total: 40 Marks